Project Documentation: Shopping List Tracker App

1. Project Overview

Project Title: Shopping List Tracker

Platform Used: MIT App Inventor

Database Used: TinyDB (Local Storage)

Brief Description:
The Shopping List Tracker is a simple mobile application developed using MIT App Inventor. The app allows users to add, view, and delete shopping items. All the items entered by the user are stored locally using TinyDB so that the list remains saved even after closing the app.

2. Objectives of the Project

- To create a user-friendly shopping list application.

- To understand how local data storage works using TinyDB.

- To implement list handling using App Inventor blocks.

- To practice event-driven programming concepts.

3. User Interface Components Used

- TextBox: To enter shopping item names.

- Button (Add): To add items to the list.

- ListView: To display the list of shopping items.

- Button (Delete/Clear): To remove selected items or clear the list.

- TinyDB: To store and retrieve shopping data locally.

4. Working of the Application

1. The user types an item name into the TextBox.
2. When the 'Add' button is clicked, the item is added to a list.
3. The updated list is stored in TinyDB using a specific tag.
4. The ListView displays all stored items.
5. When the app starts, it retrieves the stored list from TinyDB and displays it.
6. The user can select an item to delete it from the list.

5. Blocks and Logic Used

- Event Blocks (Button.Click, Screen.Initialize)

- List Blocks (Add items to list, Remove item from list)

- TinyDB Blocks (StoreValue, GetValue)

- Conditional Blocks (if statements) to check empty input.

- Variables to maintain and update the shopping list.

6. Learnings from the Project

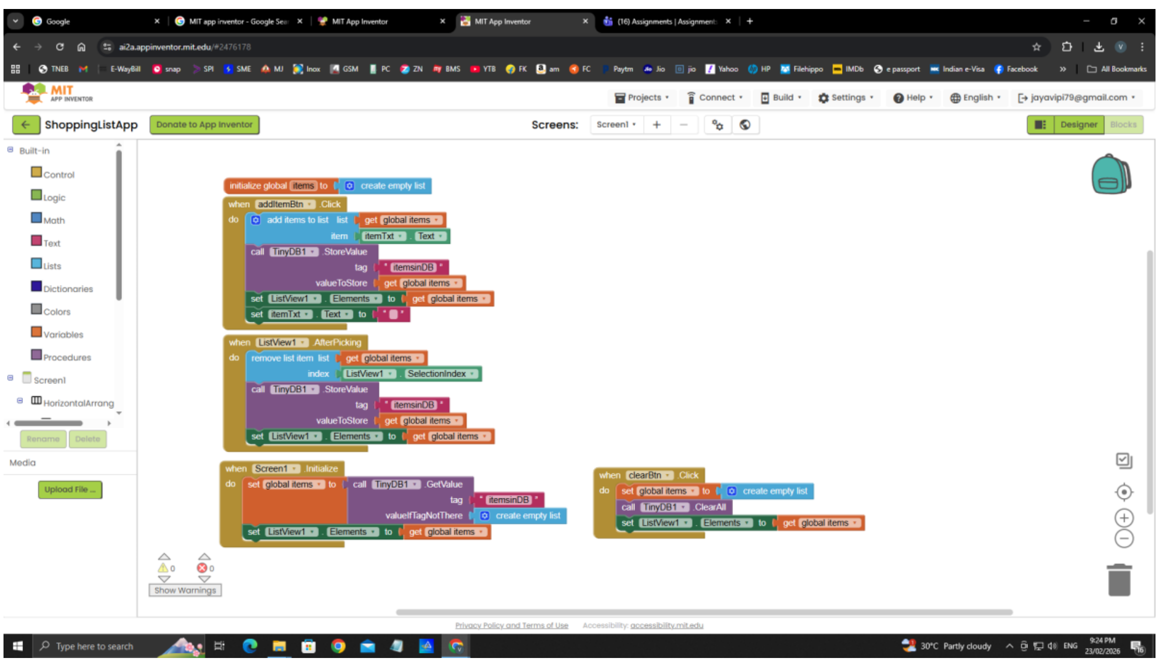
Through this project, I learned how mobile applications can store data locally using TinyDB. I understood how to use lists to manage multiple items dynamically.

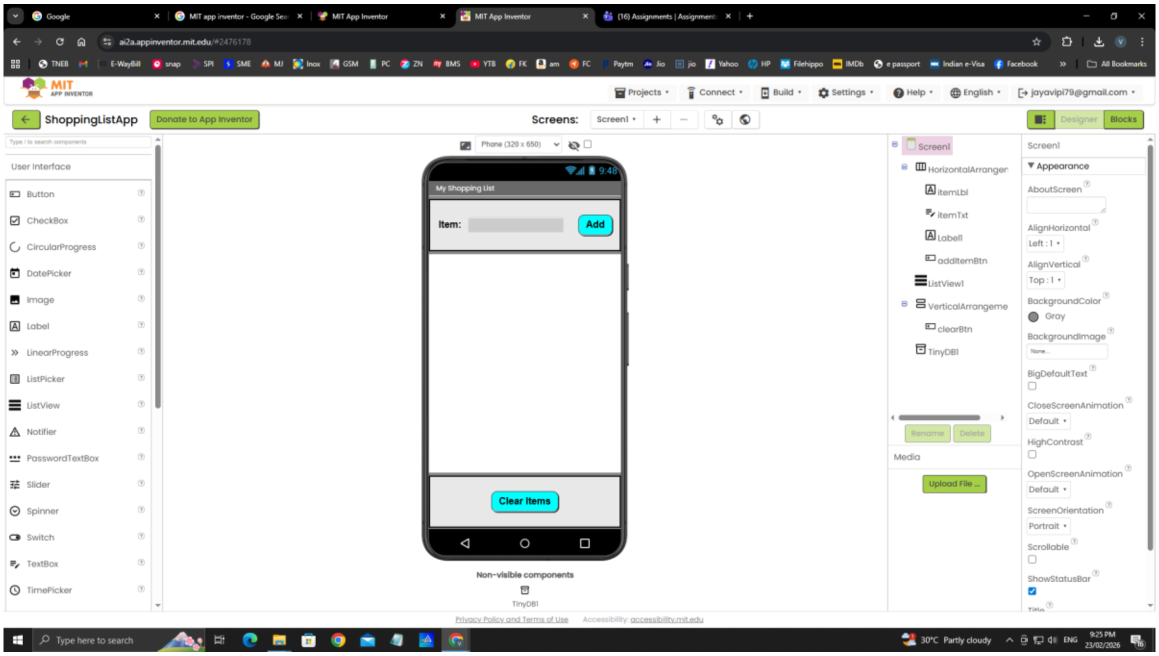
I learned how event-driven programming works, where actions happen when buttons are clicked or when the screen initializes.

This project improved my understanding of data persistence, logical thinking, and debugging errors in block-based programming.

Overall, I gained practical experience in designing a simple but functional mobile application.

Screen shots:




ShoppingListApp
Donate to App Inventor
Screens:
Screen1
Designer
Blocks
User Interface
Button
CheckBox
CircularProgress
DatePicker
Image
Label
LinearProgress
ListPicker
ListView
Notifier
PasswordTextBox
Slider
Spinner
Switch
TextBox
TimePicker
My Shopping List
Item:
Add
Clear Items
Non-visible components
TinyDB1
Screen1
HorizontalArranger
itemLbl
itemTxt
Label1
addItemBtn
ListView1
VerticalArrangeme
clearBtn
TinyDB1
Rename
Delete
Media
Upload File ...
Appearance
AboutScreen
AlignHorizontal
Left : 1
AlignVertical
Top : 1
BackgroundColor
Gray
BackgroundImage
BigDefaultText
CloseScreenAnimation
Default
HighContrast
OpenScreenAnimation
Default
ScreenOrientation
Portrait
Scrollable
ShowStatusBar
Type here to search
30°C Partly cloudy